

Designed by Remus Teodorescu 2026

LECTURE 8.1

Stationary Heat

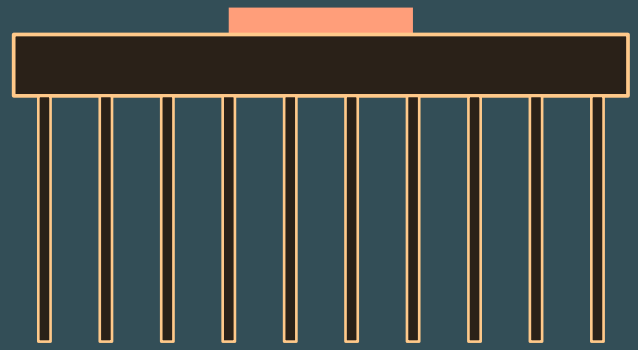
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Deep Learning for Engineering · 2026

CONDUCTION · FINS · THE HEAT SINK



Steady conduction is Poisson's equation with units: Fourier's law, a conductivity, and walls with a temperature, a flux or a convection. A power module on a finned heat sink is solved exactly, by FDM, and by a PINN that starts from the hand calculation and keeps the heat balanced.

Learning Objectives

SECTIONS, IN ORDER

1. **Heat transfer** — what heat is, its three mechanisms, which L8 solves
2. **Conduction** — the steady equation, the thermal walls, uniqueness, scaling
3. **A heat sink** — fins as one coefficient, spreading, the exact field
4. **Solving it** — FDM, a PINN told what is known, the heat balanced

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	how does heat move?
Liu ch. 2.4	what does a thermal problem need?
Incropera ch. 3	how do fins and spreading work?
Liu ch. 4.6	how is the answer checked?

Recap of L7.2 Fundamental PDEs

1 – 2 · STEADY EQUATIONS

1 **How does the type of a second-order equation decide what a network must be given before it is trained?**

An elliptic equation such as Poisson's has no time: the network takes the position only and needs a value or a flux on every wall. A parabolic one such as the heat equation takes time as a further input and needs one initial field; a hyperbolic one such as the wave equation needs two, the initial field and the initial velocity. With a condition missing the loss can still reach zero, but at one of many fields: a panel trained without its initial velocity never moves. *Liu ch. 2; Ex_07.2 notebooks 1 – 3*

2 **Why does a network trained on the Helmholtz equation tend to return a zero field, and how is that prevented?**

A zero field satisfies the Helmholtz equation and its walls for every eigenvalue with a residual of exactly zero, so gradient descent finds it first: it is the expected outcome, not a rare failure. A normalisation term in the loss fixes the scale of the mode, and a floor keeps the trainable eigenvalue away from zero. Each further mode is asked to be orthogonal to those already found, so that a new run finds the next mode instead of the lowest again. *Liu ch. 6.2, 6.4 – 6.5*

3 – 4 · EQUATIONS IN TIME

3 **How does a network represent a field in time, and why is an early error less harmful for heat than for a wave?**

Time becomes one more input, scaled by the length of the window, and the network returns the field at every position and instant at once. In the heat equation every feature decays, so an error made early is damped. A wave travels at a finite speed and keeps its shape, so an early error is carried along: in Ex_07.2 the errors of both methods fall with time on the die and grow on the panel. Finite differences met the agreed accuracy on all three problems in milliseconds. *Liu ch. 4.1, 5.1; Ex_07.2 notebooks 1, 3*

4 **How are the initial conditions given to a network in time, and why must the residual of the wave equation be scaled?**

A condition with a formula is built into the trial function, so the walls and the initial field hold for any weights. The panel's initial velocity cannot be built in the same way, so it is a term in the loss at points at the start. The panel's acceleration is several hundred times its displacement over the window, so the residual is divided by its scale; otherwise it outweighs the velocity term, and the network settles on the one motion that satisfies the rest: none. *Liu ch. 4.2, 5.2; Ex_07.2 notebooks 1, 3*

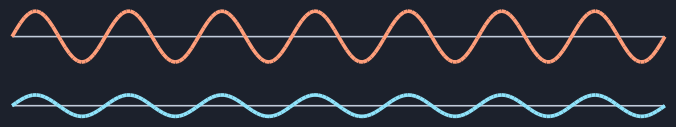
Heat

Heat — energy that flows because of a temperature difference. In a solid each atom vibrates about a fixed place; the temperature at a point is the average energy of that vibration. A hot atom shakes its neighbours through the bonds: the energy moves on, the atoms stay.

AT THE SCALE OF ATOMS

- **temperature** — at a point, an average over many atoms
- **frequency** — about 10^{13} Hz; heating widens the swing
- **bonds** — electric forces pass the push on, no radiation
- **metals** — free electrons carry heat as well

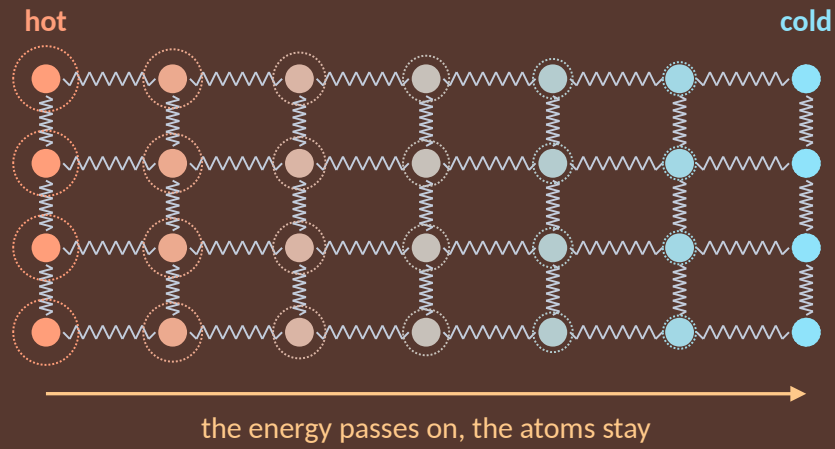
ONE ATOM AGAINST TIME



hot: a wide swing

cold: a narrow swing

ATOMS IN A SOLID, VIBRATING



Each dot is an atom and each spring a bond. The dotted ring is how far the atom swings: wide where it is hot, narrow where it is cold.

Transport Analogies

Transport analogy — heat, charge, magnetic flux, a fluid and a diffusing species all flow the same way: a difference drives a flux, a property of the material sets it, and the path opposes it with a resistance. The difference equals the resistance times the flow, an Ohm's law.

NOT A FLOW: A STRETCHED BAR

$\delta = F{\cdot}L/(E{\cdot}A)$ has the same form, but the bar stores energy, like a capacitor.

SIX FLOWS, ONE LAW

	heat	current	magnetic flux	fluid in a pipe	diffusion	charge transfer
driven by	ΔT [K]	ΔV [V]	NI [A]	Δp [Pa]	Δc [mol/m³]	η [V]
flux	q [W/m²]	J [A/m²]	B [T]	u [m/s]	j [mol/(m²s)]	i_n [A/m²]
law	$q = -k\nabla T$	$J = -\sigma\nabla V$	$B = \mu H$	$u = -(r^2/8\rho\nu)\nabla p$	$j = -D\nabla c$	$i_n \approx i_0\, \eta/V_T$
resistance	$R_{th} = L/(k{\cdot}A)$	$R = L/(\sigma{\cdot}A)$	$\mathcal{R} = L/(\mu{\cdot}A)$	$R_h = 8\rho\nu L/(\pi r^4)$	$R_D = L/(D{\cdot}A)$	$R_{ct} = V_T/(i_0{\cdot}A)$
Ohm's law	$\Delta T = R_{th}\, P$	$\Delta V = R\, I$	$NI = \mathcal{R}\, \Phi$	$\Delta p = R_h\, Q$	$\Delta c = R_D\, I/F$	$\eta = R_{ct}\, I$
unit of R	[K/W]	[Ω]	[A/Wb]	[Pa·s/m³]	[s/m³]	[Ω]
mass carried	none	none, net	none	yes, the fluid	yes, the species	yes, the ions
discussed in	L8.1 Stationary Heat L8.2 Transient Heat	L10.1 Ionic Diffusion & Charge Conservation	not in this course	L9.1 Laminar Flow L9.2 Gas Flow	L10.1 Ionic Diffusion & Charge Conservation	L10.1 Ionic Diffusion & Charge Conservation

k, σ, μ, D thermal and electrical conductivity, permeability, diffusivity, ρ, ν density, kinematic viscosity, c concentration, η overpotential, i_0 exchange current density
L, A, r length, cross-section, radius of the path, u mean speed, P, I, Φ, Q the total flows, F Faraday's constant, $V_T = RT/F$ the thermal voltage, 25.7 mV at 25 °C

Conduction

Conduction — heat carried through a material by its atoms and electrons, with no motion of the material itself. It flows from hot to cold, at a rate proportional to the temperature gradient: Fourier's law, with the conductivity k as the property of the material.

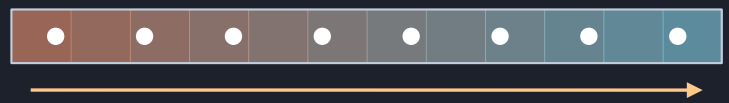
FOURIER'S LAW

$$\mathbf{q} = -k\,\nabla T$$

q the heat flux [W/m^2], k the conductivity [W/(m·K)], ∇T the gradient [K/m]

- **driven by** — a temperature gradient, inside a solid or a still fluid
- **direction** — from hot to cold, hence the minus sign
- **examples** — a chip into its heat sink, a winding into its slot

CONDUCTION, DRAWN



the solid, hot to cold

q runs down the gradient

CONDUCTIVITY OF COMMON MATERIALS

material	k [W/(m·K)]
copper	400
aluminium	237
silicon	150
stainless steel	15
glass	1
water	0.61
air	0.03

Typical values at room temperature. Four orders of magnitude lie between copper and air: a heat sink is metal, and insulation is trapped air.

Convection

Convection — heat carried away from a surface by a moving fluid. The motion of the fluid is not solved for: its effect is put into one coefficient h, and the flux is proportional to the difference between the surface and the fluid. That is Newton's law of cooling.

NEWTON'S LAW OF COOLING

$$q = h\,(T_s - T_\infty)$$

h the heat transfer coefficient [W/(m²K)], T_s the surface, T_∞ the fluid far from it

FROM THE SPEED TO THE COEFFICIENT

$$\mathrm{Re} = uL/\nu, \quad \mathrm{Nu} = C\,\mathrm{Re}^m\,\mathrm{Pr}^{1/3}, \quad h = \mathrm{Nu}\,k_f/L$$

Re, Nu, Pr the Reynolds, Nusselt and Prandtl numbers, without units, $\mathrm{Pr} = \nu/\alpha$; C, m set by the flow
h [W/(m²K)], L the length of the wall along the flow [m], u the fluid's speed [m/s]
k_f the fluid's conductivity [W/(m·K)], ν, α its kinematic viscosity and thermal diffusivity [m²/s]

CONVECTION, DRAWN



the fluid, at T_∞, moves past the surface, at T_s, gives up q

HEAT TRANSFER COEFFICIENT

flow	h [W/(m²K)]
natural, air	2 – 25
forced, air	25 – 250
natural, water	50 – 1 000
forced, water	100 – 20 000
boiling water	2 500 – 100 000

Air at 2 m/s along a 0.1 m fin: Re = 12 500, Nu = 66, h = 17 W/(m²K). At 4 m/s, h = 24 W/(m²K). Laminar flow (C = 0.664, m = 1/2) gives $h \propto u^{0.5}$; turbulent flow (C = 0.037, m = 4/5) gives $h \propto u^{0.8}$.

Radiation

Radiation — heat exchanged as electromagnetic waves between surfaces that see each other; it needs no material in between. The power a surface radiates grows with the fourth power of its absolute temperature: the Stefan–Boltzmann law, with the emissivity ε of the surface.

STEFAN–BOLTZMANN LAW

$$q = \varepsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

ε the emissivity, 0 to 1, σ the Stefan-Boltzmann constant, 5.67×10^{-8} [W/(m^{2}K^4)]
q the heat flux [W/m^2], T_s, T_{sur} the surface and the surroundings [K]

- **needs** — no medium: it crosses a vacuum
- **nonlinear** — the fourth power, with T in kelvin
- **examples** — the sun, a furnace wall, a radiator in space

RADIATION, DRAWN



the hot surface radiates
the surroundings radiate back

EMISSIVITY OF COMMON SURFACES

surface	ε
polished aluminium	0.05
anodised aluminium	0.80
black paint	0.95
a black body	1

A black body, $\varepsilon = 1$, absorbs all the radiation that falls on it and radiates the most any surface can, σT^4; a real surface radiates the fraction ε of that. At 80 °C with $\varepsilon = 0.90$, in surroundings at 25 °C, it radiates 390.50 W/m², of one size with natural convection (550 W/m² at $h = 10$ W/(m²K)).

Heat Into a Solid

Heat into a solid — heat from outside reaches the surface by contact, through a fluid or as radiation; inside the solid it is transferred by conduction alone. So conduction is the field equation a PINN solves, and convection and radiation enter as boundary conditions.

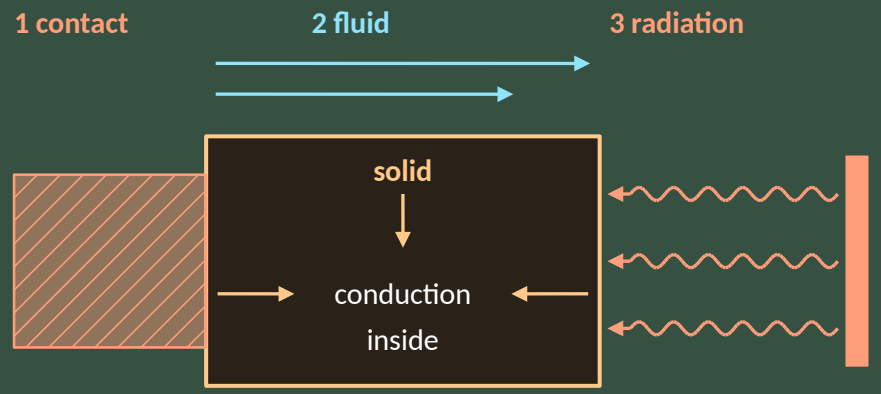
WHERE EACH MECHANISM ENTERS

	acts	in the PINN
conduction	inside the solid	the equation the PINN solves
convection	at a surface, into a fluid	a Robin wall, with h given
radiation	between surfaces	left out, or counted in h

WHY THE OTHER TWO ARE NOT SOLVED

- **convection** — its field is the flow of the fluid: that is L9
- **radiation** — T^4 and the view between surfaces: nonlinear
- **both** — act at the wall, so a boundary condition carries them

HEAT FROM OUTSIDE, THREE ROUTES



Every route ends at the surface: radiation is absorbed in a thin layer there. Inside, it is transferred by conduction. Radiation is not small: at 80 °C against 25 °C it acts like h = 7.10 W/(m²K).

Steady Conduction

Steady conduction — Fourier's law in an energy balance: in steady state what leaves a small volume equals what is generated in it, $\nabla\cdot\mathbf{q} = Q$. With $\mathbf{q} = -k\nabla T$ that is Poisson's equation for the temperature, with Q/k as its source.

FOURIER'S LAW IN AN ENERGY BALANCE

$$\mathbf{q} = -k\,\nabla T, \qquad \nabla\cdot\mathbf{q} = Q$$

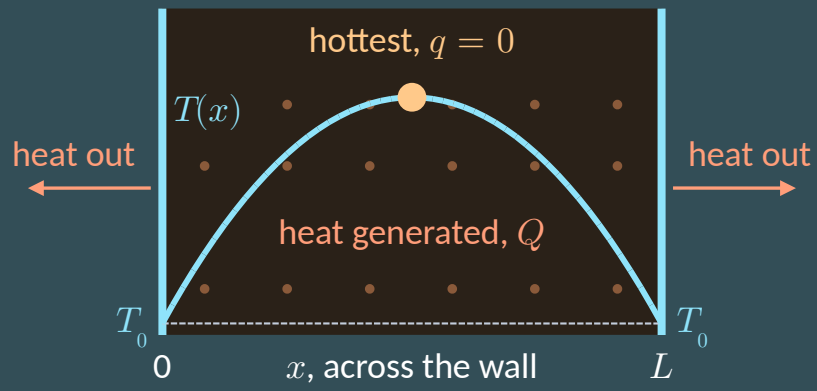
$$T_{xx} + T_{yy} + \frac{Q}{k} = 0$$

$\mathbf{q}$ the heat flux, k the conductivity [W/(m·K)]
Q the heat generated [W/m^3]

CONDUCTION IN PYTORCH

```
T = net(xy)                          # the temperature, in kelvin
q = -k * grad(T, xy)                 # Fourier's law; grad = autograd
F = div(q, xy) - Q                   # the energy balance, zero inside
```

A WALL THAT GENERATES HEAT



$$T(x) = T_0 + \frac{Q\,x\,(L-x)}{2k}, \qquad T_{\max} = T_0 + \frac{QL^2}{8k}$$

Why a parabola: the heat crossing a plane at x is all that is generated between it and the middle, so the flux and the slope of T grow linearly from there. Aluminium at 1 MW/m^3 and 100 mm thick, rises 7.49 K.

Thermal Boundary Conditions

Thermal boundary conditions — three ways to fix a wall: a temperature (Dirichlet), a heat flux (Neumann), or convection to a fluid (Robin), which ties the flux to the wall temperature. Only a temperature can be built into the network; the other two stay loss terms.

A TEMPERATURE, A FLUX, A CONVECTION

Dirichlet

$$T = \bar{T} \text{ on } \partial\Omega_D$$

Neumann

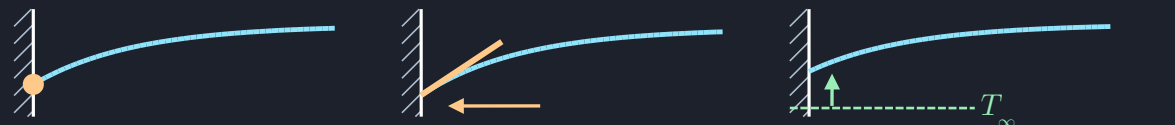
$$-k\frac{\partial T}{\partial n} = \bar{q} \text{ on } \partial\Omega_N$$

Robin

$$-k\frac{\partial T}{\partial n} = h\,(T - T_\infty)$$

$\bar{T}$ the wall temperature, $\bar{q}$ the wall heat flux, n the outward normal
h the heat transfer coefficient, T_∞ the fluid temperature

WHAT EACH CONDITION FIXES AT THE WALL



Dirichlet the value — **Neumann** the slope — **Robin** slope tied to value

ROBIN · CONVECTION AT THE WALL

- **balance** — heat conducted to the wall equals heat the fluid carries away
- h — heat transfer coefficient [W/(m²K)], or film coefficient: the fluid layer at the wall in one number

BIOT NUMBER

$$\mathrm{Bi} = \frac{h\,L}{k} = \frac{L/k}{1/h}$$

h [W/(m²K)], L the size of the body [m], k its conductivity [W/(m·K)]

The heat meets two resistances in series: the body, L/k; then the fluid at its surface, $1/h$. Bi says where the temperature drops. Bi ≪ 1: in the fluid; the body is nearly uniform, the wall nearly insulated (Neumann). Bi ≫ 1: in the body; the surface sits at T_∞ (Dirichlet).

Unique Solution

Unique solution — a problem has one when a single field fits its equation and its walls. With a flux on every wall steady conduction has many: a flux fixes the slope of T at a wall, never its level, so $T + c$ fits the same equation and walls for any constant c.

A FLUX FIXES THE SLOPE, NOT THE LEVEL

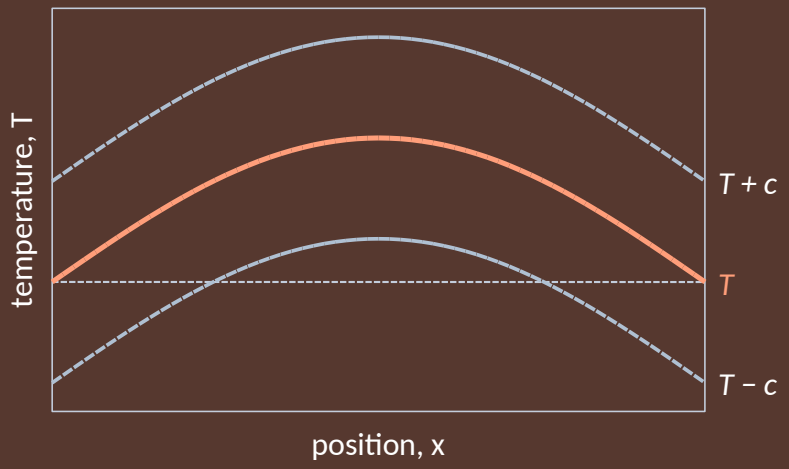
$$\nabla^2(T+c) = \nabla^2 T, \qquad \frac{\partial(T+c)}{\partial n} = \frac{\partial T}{\partial n}$$

c any constant, *n* the outward normal of a wall

- **a temperature or a convection wall** — fixes the level
- **a flux on every wall** — pin one value in the loss

TALK · 2 MIN Why does a flux on every wall not fix the temperature, in your own words? Then, in the plot on the right, which of the three curves is right, and what would you add to the loss to choose it?

THE SAME FIELD, SHIFTED



Same curvature, same slopes at the walls: the equation and every flux condition fit all three. With flux on every wall the heat out must also equal the heat generated, the balance of Steady Conduction.

Non-Dimensional Form

Non-dimensional form — the problem rewritten in scaled variables: lengths divided by the device size L, temperature by a rise $\Delta T = QL^2/k$. Every term is then of order one, as the initialisation and the loss weights assume. Train scaled, and convert back at the end.

SCALE, TRAIN, CONVERT BACK

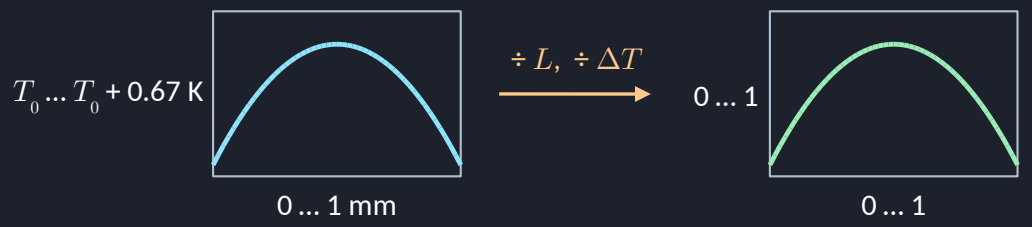
$$\theta = \frac{T - T_0}{\Delta T}, \quad \xi = \frac{x}{L}, \quad \Delta T = \frac{Q L^2}{k}$$

θ the scaled temperature, ξ the scaled coordinate, ΔT the temperature scale

$$\theta_{\xi\xi} + \theta_{\eta\eta} + 1 = 0$$

$\eta = y/L$; the same equation, every coefficient equal to one

THE SAME CURVE, SCALED



A SILICON DEVICE, BEFORE AND AFTER

	physical	scaled
length L [mm]	1	1
conductivity k [W/(m·K)]	150	1
source Q [W/m³]	10^8	1
rise $\Delta T = QL^2/k$ [K]	0.67	1
coefficient h [W/(m²K)]	25	$Bi = 1.67 \times 10^{-4}$

In metres and kelvin the source term is ten to the eighth and the length ten to the minus third. Scaled, every term is of order one.

Fin Efficiency

Fin efficiency — the heat a fin gives to the air over what it would give if all of it were at its root's temperature. A fin is conduction along it and convection from its faces, with an exact solution, so a row of fins acts on the base as one effective coefficient h_{eff}.

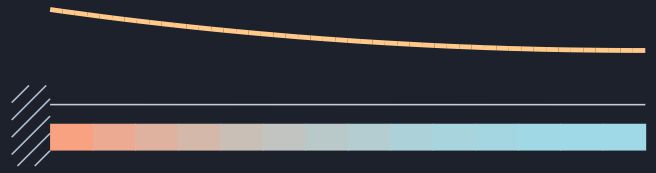
ONE FIN EXACTLY, THEN THE ROW AS ONE COEFFICIENT

$$\eta = \frac{\tanh(mL_c)}{mL_c}, \quad m = \sqrt{\frac{2h}{k\,t_f}}, \quad L_c = L_f + \frac{t_f}{2}$$

$$h_{\text{eff}} = h\,\frac{\eta\,2L_c + p - t_f}{p}$$

t_f, L_f the fin's thickness and length, L_c its length with the tip
p the pitch, h the air's coefficient, η the fin's efficiency

ALONG ONE FIN, ROOT TO TIP

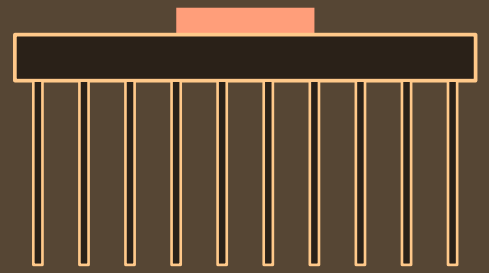


the root, at the base: hottest

the tip: 83 % of the root's rise

so η = 0.88

THE HEAT SINK OF EX_08



	value
fin efficiency η	0.884
h_{eff} on the base	322.1 W/(m²K)
against the bare h	8.1 times

Typical values. Mini project MP8.1A checks the collapse.

Heat Spreading

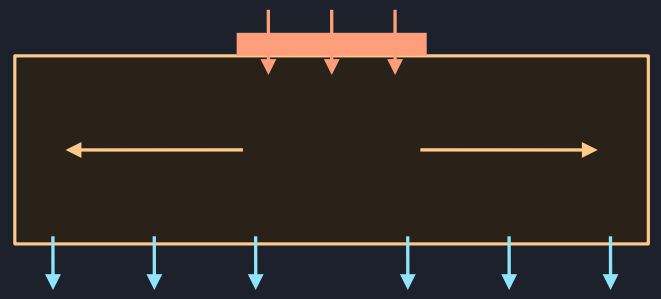
Heat spreading — heat that enters a plate on a small area and leaves it over a large one has to run sideways first, so the plate is hotter under the source than its mean. A one-dimensional hand calculation spreads the heat over the whole base at once and misses exactly that.

THE HAND CALCULATION, IN ONE DIMENSION

$$T_{\text{fin}} = T_\infty + \frac{q_{\text{mean}}}{h_{\text{eff}}}, \qquad T_{\text{top}} = T_{\text{fin}} + \frac{q_{\text{mean}}\, t_b}{k}$$

q_{mean} the module's heat spread over the whole base
t_b the base's thickness, T_∞ the air

THROUGH THE BASE PLATE

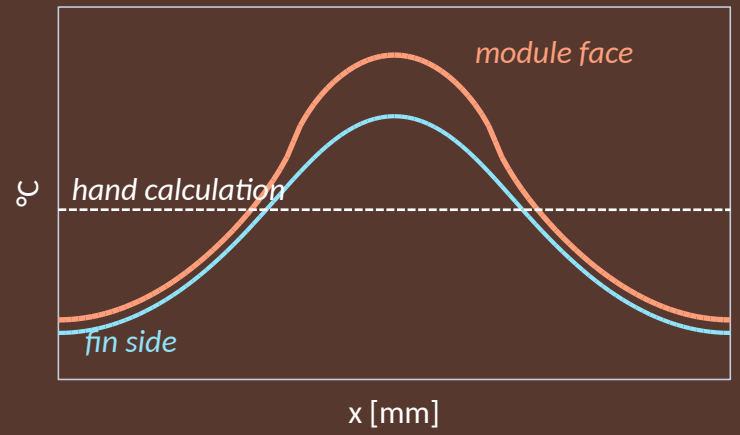


in: on 30 mm, 50 kW/m²

sideways: the spreading

out: over 100 mm, to the fins

THE BASE, ALONG ITS WIDTH, EXACT



The exact hottest point is 92.45 °C; the hand calculation says 87.47 °C. The spreading adds 4.98 K, a tenth of the rise.

Exact Solution

Exact solution — with insulated ends, cosines across the base solve the equation exactly, and each cosine has its own profile through the thickness in elementary functions. Their sum is the temperature to rounding, so FDM and the PINN can be scored against the truth itself.

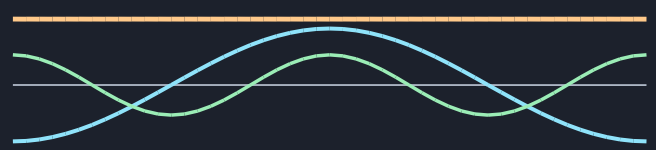
A COSINE SERIES, EACH TERM ITS OWN PROFILE

$$\theta(x,z) = \sum_{n \geq 0} \cos\left(\frac{n\pi x}{W}\right) Z_n(z)$$

$$Z_n = \frac{q_n}{k\lambda_n} \frac{\cosh \lambda_n z + r_n \sinh \lambda_n z}{\sinh \lambda_n t_b + r_n \cosh \lambda_n t_b}$$

$\lambda_n = n\pi/W$, $r_n = h_{\text{eff}}/(k\lambda_n)$, W the base's width
q_n the cosine coefficients of the module's flux

THE FIRST TERMS, ACROSS THE BASE

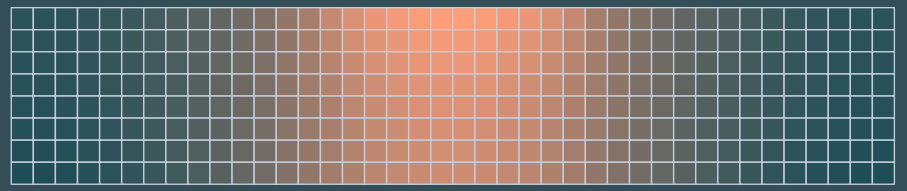


n = 0: the mean, the hand calculation

$n \geq 1$: the spreading

each fades through the thickness

THE BASE PLATE, EXACT



the height drawn twice its scale

83.5 °C — 92.4 °C

hottest point	92.45 °C, under the centre
heat out	150.00 W of 150 W
the slow part	summed in closed form

FDM on a Rectangle

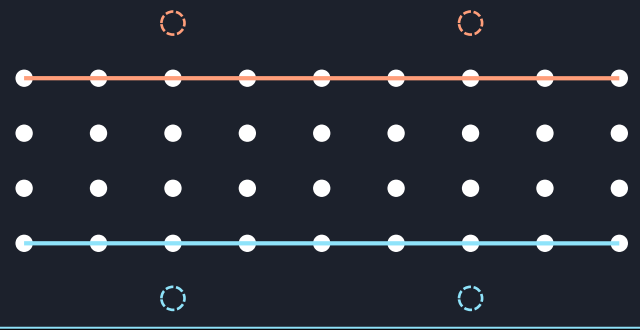
Finite differences (FDM) — the derivatives replaced by differences between neighbouring nodes of a grid, so the equation becomes one linear system. On a rectangle every wall lies on a grid line; a flux or a convection is imposed through a ghost node one spacing outside the face.

THE TWO FACES, THROUGH GHOST NODES

$$\theta_{-1} = \theta_1 - 2h_z\,\mathrm{Bi}\,\theta_0, \qquad \theta_{N+1} = \theta_{N-1} + 2h_z\,\mathrm{Bi}\,g$$

θ_{-1}, θ_{N+1} ghost nodes outside the two faces, h_z the spacing across
$g = q''/q_{\mathrm{mean}}$ the scaled flux, $\mathrm{Bi} = h_{\mathrm{eff}} t_b / k$

THE GRID ON THE BASE, SCALED



a ghost above: the module's flux

the nodes: every wall on a grid line

a ghost below: the convection

REFINED AGAINST THE EXACT SOLUTION

grid	worst error	
21 × 5	3.73e-02 K	
41 × 9	1.85e-02 K	
81 × 17	9.16e-03 K	meets 0.01 K
161 × 33	4.54e-03 K	

The error halves with each halving of the spacing and sits at the footprint's edges, where the flux steps: first order there, second elsewhere. One solve of 1377 unknowns takes a few milliseconds.

Built-In Hand Calculation

Built-in hand calculation — the trial function starts from what the engineer already knows, here the one-dimensional hand calculation $1 + \mathrm{Bi}\cdot Z$, and the network learns only what it misses, the spreading. Taking $\cos(\pi X)$ as its input makes both insulated ends hold for any weights.

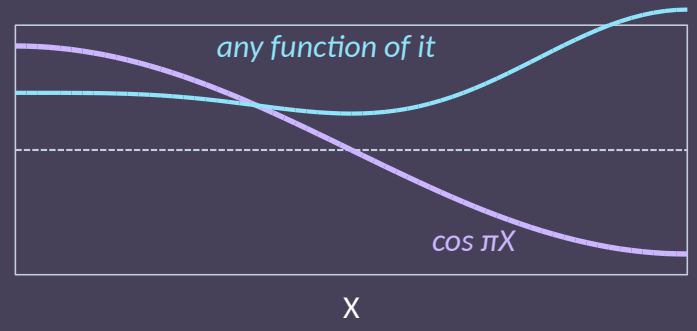
THE KNOWN PART, PLUS A SMALL LEARNT ONE

$$\hat{\theta}(X, Z) = 1 + \mathrm{Bi}\, Z + S\, \mathcal{N}(\cos \pi X,\ Z)$$

$1 + \mathrm{Bi}\, Z$ the hand calculation, $S = 0.1$ the size of the spreading
$\mathcal{N}$ the network, $X = x/W$, $Z = z/t_b$

TALK · 2 MIN How would you explain, in your own words, why a PINN should start from the hand calculation? Then, in the plot on the right, why does any function of cos πX have zero slope at both ends?

THE ENDS, BY THE CHOICE OF INPUT



THE TRIAL FUNCTION

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)  # 2 in
f = torch.cat([torch.cos(pi*X), Z], 1)      # even at the ends
theta = 1 + Bi*Z + 0.1*net(f)               # hand calc + rest
```

Energy Balance Loss

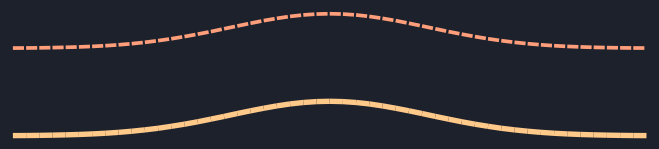
Energy balance loss — a loss term that asks the heat leaving through the fins to equal the heat the module puts in. No wall of the base holds a temperature, so the level of the field is held only weakly by the convection; the balance pins it.

THE RESIDUAL, THE TWO FACES, AND THE BALANCE

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + w_F\,\mathcal{L}_{\text{faces}} + w_E\left(\overline{\hat{\theta}(X,0)} - 1\right)^2$$

$\overline{\hat{\theta}(X,0)}$ the fin side mean : 1 when the heat out is the heat in
w_F, w_E the weights of the faces and of the balance

THE FIN SIDE, ACROSS THE BASE



without the balance: floats

the same shape, a kelvin off

schematic

with it: on the exact level

FDM AND PINN, SCORED EXACTLY

	FDM, 81 × 17	PINN
hottest point	92.45 °C	92.47 °C
worst error	0.01 K	0.22 K
one answer	3.6 ms	1.0 ms
offline	none	407 s training
heat out	150.00 W	150.05 W

Ex_08 section 5, on a laptop GPU; on Colab the times differ. On a rectangle FDM is the more exact; the PINN's case is many answers, which Part 2 and the mini projects take up.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
level set and multiplier	a curved wall, built in	Liu ch. 3	a plate with a bore
fitted-mesh reference	FEM where no series exists	Liu ch. 2.10	shapes with no exact answer
variable conductivity	$k(\mathbf{x})$ kept inside the divergence	Liu ch. 2.4	layered or composite walls
manufactured solution	a chosen answer, so an error curve	Liu ch. 2.4	testing a new solver
the fins resolved	conduction in the whole comb	Incropera ch. 3	mini project MP8.1A
energy method	an energy, first derivatives only	Liu ch. 2.8	elasticity, structures
diffusion surrogate	trained on solved cases, then steered	UDL ch. 14 – 18	many boundary conditions
PINN or FEM	when each is the better tool	Liu ch. 2.10	one-off steady solves

WHY THEY ARE LEFT OUT

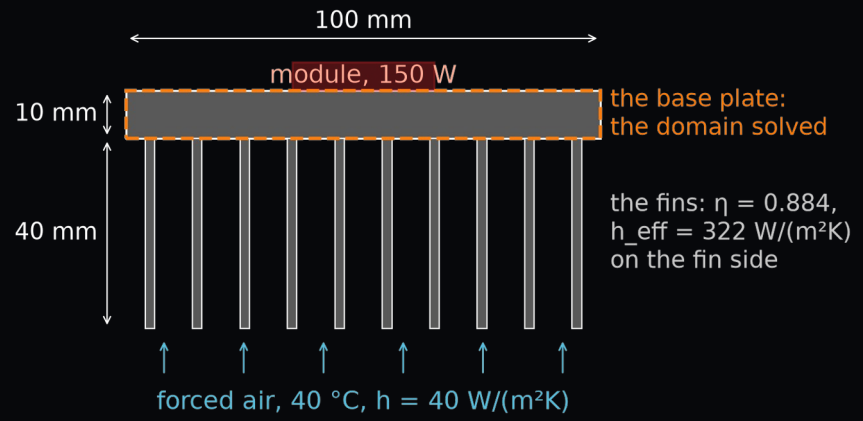
Ex_08's base is a rectangle of one material with an exact solution, so it needs no curved wall, no reference and no manufactured answer. The first two rows were this lecture's while its exercise was a plate with a hole.

EXERCISE

Exercise 8 Part 1

ONE NOTEBOOK, PART 1 SECTION BY SECTION

- **1 – 3** — the heat sink, its fins, the exact solution
- **4** — FDM, refined until it meets 0.01 K
- **5** — a PINN with the hand calculation built in
- **6** — what the comparison says
- **Part 2** — the same sink switched on, in L8.2



THE QUESTION EACH PART ANSWERS

1 – 3	how hot is the module?
4	how fine must the grid be?
5	what must the PINN be told?

REPORT QUESTION, TO CONCLUDE

For the steady heat sink solved once, which method would you choose, and in which use - many footprints, many powers, or an unknown coefficient - would you choose the PINN instead?

SUMMARY

Key Takeaways

HEAT TRANSFER

what it is	conduction, convection, radiation
function	T by conduction; the rest are walls
train, tune	scale first; Bi says which wall
vs Poisson	units, three walls, a level to fix
used for	devices, busbars, heat sinks

A HEAT SINK, SOLVED

what it is	fins as h_{eff} on a base plate
function	the base's temperature, exact
train, tune	hand calculation in; balance in loss
vs FDM	less exact; as fast once trained
used for	many sinks, powers, footprints

Next — L8.2: the same heat sink, switched on for twenty seconds.

Questions

1 – 2 · CONDUCTION FOR A NETWORK

1 **Which heat transfer does a network solve as a field equation, and how do convection and radiation enter its training?**

Inside the solid heat is transferred by conduction alone, so steady conduction is the field equation the network is trained on, at points inside the base. Convection and radiation act only at the surface and enter as boundary conditions: a convection wall ties the heat flux to the wall temperature, and radiation is left out or counted in the heat transfer coefficient. Only a fixed temperature can be built into the network; a flux or a convection stays a loss term. *Incropera ch. 1, 3; Ex_08 sections 1 – 3*

2 **Why must the coordinates, the temperature and the loss terms be scaled before a network is trained on the heat sink?**

In metres and kelvin the loss terms differ by orders of magnitude, while the initialisation of the network and the loss weights assume terms of order one. In Ex_08 the position is divided by the width and the thickness of the base, and the temperature rise by the hand calculation's own rise. The residual and the conditions on the two faces are also divided by the small Biot number, so that every term of the loss is of order one. *Liu ch. 4.4; Ex_08 sections 3, 5*

3 – 4 · THE HEAT SINK'S NETWORK

3 **What does the heat sink's network have built in, and what is left for it to learn?**

The trial function starts from the one-dimensional hand calculation through the thickness of the base, which holds most of the solution; the network adds only the spreading of the heat sideways under the module, scaled to a tenth. Its input is the cosine of the scaled position along the base, so both insulated ends have zero slope for any weights: hard enforcement by the choice of input. The module's heat flux and the fins' convection remain loss terms. *Liu ch. 2.2; Ex_08 section 5*

4 **Why does the heat sink's network need an energy balance in its loss, and when does it pay against finite differences?**

No wall of the base holds a temperature, so the convection fixes the level of the field only weakly, and without a further term the network drifts by about a kelvin. A loss term asks the heat leaving through the fins to equal the heat the module puts in, which pins the level within 0.22 K of the exact solution. Finite differences are more exact here, 0.01 K in milliseconds; a network would pay where one training answers many powers or footprints. *Liu ch. 4.6; Ex_08 sections 4 – 6*

NEXT

L8.2 — Transient Heat

The same heat sink, switched on for twenty seconds: storage, causality, and the fins' coefficient found from one thermocouple.

BEFORE THE NEXT SESSION

read	Incropera ch. 3
run	Ex_08, Part 1, sections 1 to 6
bring	the hottest point, and your error
think about	what you know before you train

The steady heat sink is solved, exactly and twice. Next it is switched on.